VYOM KHANNA

Senior BI Engineer | Analytics Engineer | Senior Data Analyst | Data Engineer

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PROFILE

BI Developer & Analytics Engineer with 5 years of experience, seeking a permanent role after building a diverse portfolio across sectors. Comfortable working closely with stakeholders to translate ambiguous business requirements into clear analytical scope, managing expectations throughout delivery, and communicating findings confidently to both C-suite and non-technical operational audiences. Core expertise spans data analysis, BI engineering, dimensional data modelling & ETL/ELT pipeline development, underpinned by best-practice SQL, Power BI, and data modelling standards, in compliance with UK GDPR, DPA 2018, and FCA.

TECHNICAL SKILLS

Languages: SQL, Python, DAX, Power Query
BI & Visualisation: Power BI, Qlik Sense/View, Tableau, Excel, Looker
ETL & Orchestration: Qlik, dbt, Medallion Architecture (Bronze/Silver/Gold), Qlik QMC, Task Scheduler
Databases/Cloud: Snowflake, Azure, MySQL, PostgreSQL
ML Tools: Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn
Data Modelling: Dimensional Modelling, Fact & Dimension Design, Snowflake & Star schema (Kimball)
Certificates: Databricks, dbt, Power BI, Tableau, Looker
AI Tools: Claude, Copilot
Other: Git, CI/CD, Agile/Scrum, Documentation, Amplitude, GA4

PROFESSIONAL EXPERIENCE

Analytics & Business Intelligence Engineer

Jan 2025 - Present

London, Hybrid/Remote, United Kingdom
Note: List of short-term contracts across multiple sectors

Spotlight Sports Group (Current Contract)

- **Engineered** Snowflake SQL tables & scripts to define & segment CRM audiences, enabling **reverse ETL** into **Braze** for the CRM & marketing teams to execute **win-back campaigns** & subscriber reacquisition initiatives.
- Built a subscriber, churn, onboarding & revenue performance dashboard in **Looker Studio**, underpinned by **Snowflake SQL table as a semantic layer**, serving as the leadership team's **single source of truth**, designed for **stakeholder consumption** without requiring technical interpretation.
- Developed app-based **article performance scorecard** views across multiple **content verticals**, tracking delivery progress & reporting status updates to stakeholders.
- Built **dbt models** on Snowflake with **incremental load strategies**, **staging-to-mart architecture**, ref() dependency management, and built-in **data quality tests** across subscription and transaction data.

GreenSquareAccord

- Designed a Power BI dashboard across **24,000 properties** & **15,000 repair** jobs on a **Kimball star schema** & **centralised DAX table** with postcode-level & EPC-based transformations for compliance & cost analysis.
- Surfaced **£52K** in potential annual contractor savings & flagged **7,600** EPC D-or-below properties as the primary **Decent Homes 2** compliance risk, feeding directly into **Tenant Satisfaction Measure** reporting.
- Established a **data dictionary** covering **KPI definitions**, DAX calculations, source tables, data grain & known limitations - the single reference for cross-departmental disputes & **compliance audits**.
- Refactored SQL logic into a **dbt staging** & **presentation layer**, standardising column naming, enforcing data types and deduplicating records, cutting **downstream DAX complexity** and improving refresh performance.
- Mentored non-technical business users through **Power BI adoption** & junior analysts in SQL & DAX, producing plain-language user guides & hands-on walkthroughs.

Avis Budget Finance

- Built a **machine learning** demand forecasting model for car allocation across airport locations, engineering features from weather, vehicle type & location data, evaluating **Linear Regression, Random Forest** & **XGBoost**.
- Delivered fleet distribution recommendations from model outputs, identifying key demand drivers by location & vehicle type to reduce misallocation **risk** & support operational planning.
- Presented Python **predictive model** outputs incorporating historical demand signals & weather variables to operational planning teams, enabling the fleet team to act on **forward-looking demand signals** without needing technical interpretation.
- Maintained a shared tracker for delivery visibility across **metric definition** & **sign-off** workstreams, covering dependencies & stakeholder milestones, proactively surfacing blockers.

Frasers Hospitality

- Optimised an underperforming Power BI dashboard by migrating calculated columns upstream into SQL, reducing report load time from **5 minutes to 30 seconds**.
- Transformed raw transactional data into business-ready **fact** & **conformed dimension tables**, enabling consistent KPIs & cross-functional reporting, cutting ad hoc reporting by **25%**.
- Delivered an On-The-Book vs Forecast revenue dashboard across cities, identifying a **25% revenue gap** driven entirely by **volume shortfall** & surfacing day-of-week variance indicating a corporate demand profile.
- Documented ETL flow across **Bronze/Silver/Gold** pipeline layers, enabling faster debugging, **audit-readiness** & transparent handover.

Poundland & Dealz

- Built a **loss-prevention Power BI dashboard** tracking **shrinkage** by category, time of day & department, giving management a ranked view of avoidable losses to prioritise intervention.
- Restructured store-level extracts into a centralised **dbt transformation layer**, standardising **KPI definitions** and eliminating duplicate aggregation logic across area and store reporting.
- Produced area-wide MI packs consolidating **like-for-like sales**, **labour cost %** & stock availability across the South East region (90 Stores), with report guides for non-technical staff.
- Delivered **SKU-level** performance reports analysing units per transaction, **basket behaviour** & category trends, enabling data-led ranging & shelf allocation decisions.

Oodle Car Finance

- Identified urban applicants as **65%** more likely to fund than rural, ages 36-65 representing **78%** of all funded loans, & the Urban 51-65 Convertible segment as the highest-value cohort at a **50%** funded rate.
- Audited & redesigned low-adoption Power BI dashboards, resolving **SQL performance** issues, rebuilding the visual layer & adding missing **KPIs** for daily stakeholder monitoring.
- Implemented **Python data validation** scripts covering row counts, null checks & **anomaly detection** at each pipeline stage, flagging structural nulls, fallback values & outlier ratios.

Business Intelligence Developer

Sep 2021 - Aug 2023

Team Computers Pvt Ltd (Client: Care Health Insurance), India

- Led **end-to-end delivery lifecycle** across BI projects - gathering stakeholder requirements, documenting metric definitions & data grain, managing **UAT sign-off** via Excel & Jira-based trackers, & designing **full-stack analytics** solutions using SQL, Python, dbt, Power BI, Qlik & Snowflake.
- **Built** a Claims Pendency Report, giving the claims department a daily prioritised view by complexity & value, reducing pending claims from **80K to 38K**, cutting resolution from **4 days to 3 hours**, preventing financial leakage & enabling the Sales team to use turnaround time as a proof point in renewal conversations.
- **Optimised** ETL/ELT pipeline performance by streamlining Qlik scripts, migrating Power BI calculated columns upstream into SQL, centralising DAX measures to eliminate duplicates, & redesigning **Snowflake schema to Kimball star schemas** - reducing overnight processing time from **8 hours to 5 hours** & improving dashboard refresh speeds across the organisation.
- Delivered an integrated CRM dashboard surfacing **NPS**, **churn patterns** & **renewal trends** - 70% of non-renewals traced to premium hikes & poor reminders, informing automated communications that contributed to a 20% retention uplift.
- Concurrently delivered three high-priority BI initiatives - **Sales Executive Report**, **Claims Automation** & **Commission Optimisation** - building reusable ETL pipelines & data models, delivered on time.
- Built a **four-module executive reporting** system for the **MD** across Claims, Finance, Sales & Actuarial, integrating real-time KPIs via modular SQL & DAX as a standard reference for MD reviews.
- Developed Power BI dashboards analysing customer behaviour across insurance products, reducing claim resolution time by **30%**.
- Automated **bordereaux** generation for B2B insurance partners, stripping **PII** & structuring claims data into **compliant submission formats** for external distribution via secure portal.
- Migrated **100+ reports** from Excel to Power BI/Qlik, automating delivery via Python, Qlik NPrinting & QMC - reducing manual effort by **70%** & ensuring PHI-free, IRDAI-compliant outputs.
- Developed Python predictive models to flag high-risk claims, enhancing **fraud detection** & **underwriting precision**.

Data Analyst

Aug 2020 - Aug 2021

Nipa International, India

- Streamlined ERP reporting by building a batch-based **PySpark** data pipeline to automate extraction, cleaning & visualisation preparation, reducing processing time by 20%.
- Supported the preparation of reporting datasets by structuring transactional data into **fact** & **dimension** tables for operational KPI analysis.
- Automated Excel & SQL reports to track production KPIs, efficiency, downtime & defects, for faster performance monitoring.
- Uncovered **Rs.25L (£24K)** in potential annual savings through material waste analysis.

EDUCATION

University of Southampton, UK
Sep 2023 - Sep 2024
Masters Data & Decision Analytics (Merit)

SRM Institute of Science & Technology, India
Jul 2016 - Aug 2020
Bachelors in Mechanical Engineering (Merit)